ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

MECHATRONICS

PAPER 1 · PRACTICE B · MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central NC(V) Level 4 Mechatronics paper 1 · practice b.

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How to mark: each ✓ is one mark. Give marks for the formula, the substitution and the answer with the correct unit. Accept other correct answers, sensible project decisions and neat, recognisable sketches, circuits and charts.

QUESTION 1

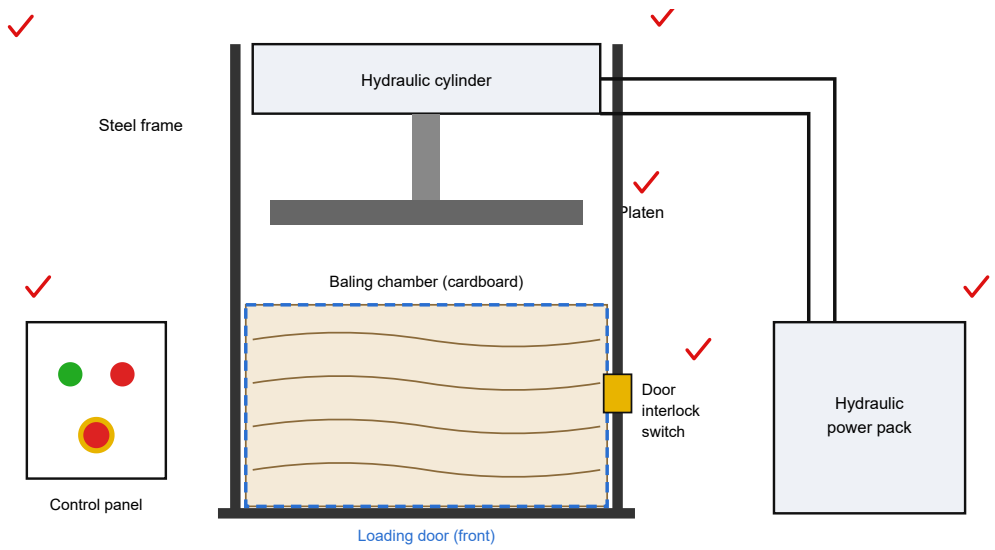
1.1 A constraint is a limit or restriction that the design must work within ✓. Any THREE ✓✓✓: budget; floor space and height of the building; available electrical supply (e.g. 380 V three-phase); safety laws (OHS Act); time to complete the project; size of the bales needed by the buyer. (4)

1.2 Any SIX sensible specifications ✓✓✓✓✓✓, e.g. bale size 800 × 600 × 1 000 mm; compression force of about 90 kN; cycle time under 60 s; door interlock and emergency stop; runs on 380 V three-phase; noise below 85 dB; cost below the budget; strong steel frame that does not bend. (6)

[10]

QUESTION 2

2.1



Neat freehand sketch ✓ showing: frame with the baling chamber and loading door ✓; interlock switch on the door ✓; cylinder with platen above the chamber ✓; power pack and control panel ✓ (5)

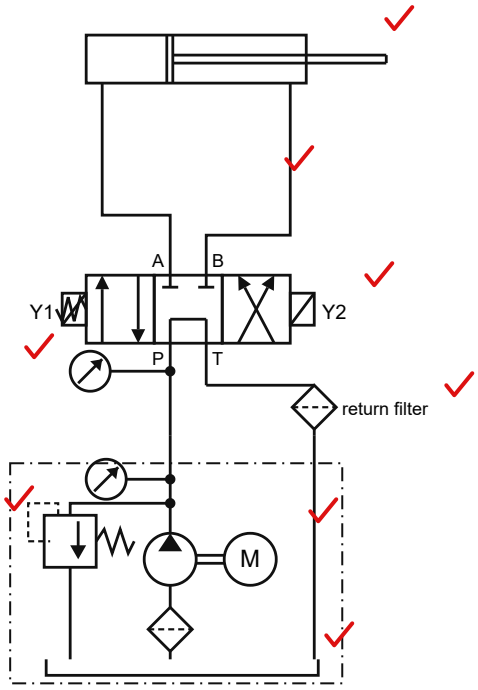
2.2.1 $A = \pi d^2 \div 4 = \pi \times 0{,}1^2 \div 4 = 7{,}854 \times 10^{-3}\ m^2$ ✓; $F = p \times A = 12 \times 10^6 \times 7{,}854 \times 10^{-3}$ ✓ $= 94\ 248\ N = 94{,}25\ kN$ ✓ (3)

2.2.2 $Q = 20 \div 60\ 000 = 3{,}333 \times 10^{-4}\ m^3/s$ ✓; $v = Q \div A = 3{,}333 \times 10^{-4} \div 7{,}854 \times 10^{-3}$ ✓ $= 0{,}042\ 4\ m/s = 42{,}4\ mm/s$ ✓ (3)

2.2.3 $A_{annulus} = \pi(0{,}1^2 - 0{,}05^2) \div 4 = 5{,}890 \times 10^{-3}\ m^2$ ✓; $v = 3{,}333 \times 10^{-4} \div 5{,}890 \times 10^{-3} = 0{,}056\ 6\ m/s = 56{,}6\ mm/s$ ✓ (2)

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2.3



Tank, filter and pump ✓; relief valve from the pump line to the tank ✓; pressure gauge ✓; 4/3-way valve with three positions ✓, tandem centre ✓, solenoids and centring springs ✓; A and B lines to the double-acting cylinder ✓; return line to the tank ✓

(8)

2.4 The stroke length changes with the amount of cardboard ✓, so the stroke must end when the bale reaches the set pressure (tightness), not at a fixed position ✓. (2)

2.5 Any TWO ✓✓: door interlock switch (press cannot move with the door open); emergency stop; two-hand control for manual moves; light curtain; pressure relief. (2)

[25]

QUESTION 3

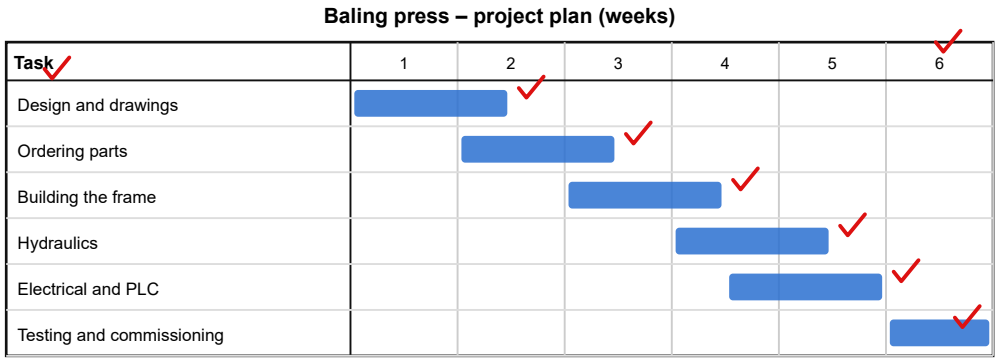
3.1

Item	Description	Quantity	Unit cost (R)	Total (R)
1	Double-acting cylinder 100/50 mm	1	9 500	9 500
2	Hydraulic power pack 20 L/min	1	18 000	18 000
3	4/3 solenoid valve (tandem centre)	1	4 200	4 200
4	Pressure switch	1	1 300	1 300
5	Door interlock switch	1	650	650
6	PLC with I/O	1	6 500	6 500
7	Steel frame and platen	1	15 000	15 000

Suitable headings ✓✓; any SIX sensible items ✓✓✓✓ (2 items per mark); totals and a grand total ✓✓

(8)

3.2

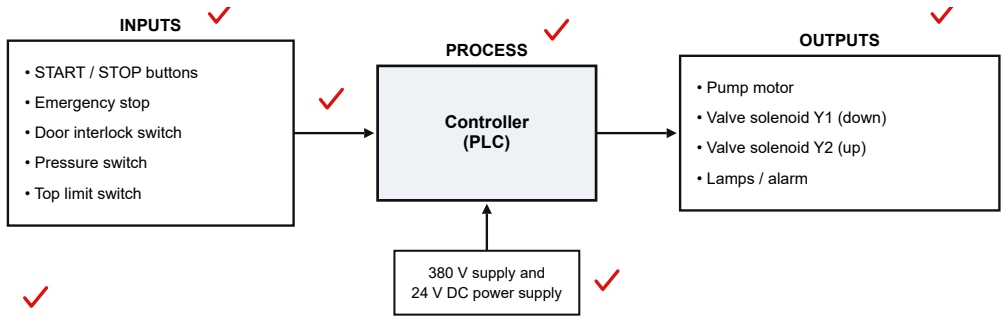


Week scale 1–6 along the top ✓; SIX sensible activities ✓✓ (e.g. design, ordering, building the frame, hydraulics, electrical and PLC, testing and commissioning); bars of sensible length in a logical order ✓✓✓✓; overlapping activities ✓; neat with a title ✓

(10)

3.3 P (proportional): gives a correction in proportion to the error, a fast response, but leaves a small steady error ✓✓. I (integral): adds up the error over time and removes the steady-state error ✓✓. D (derivative): reacts to how fast the error changes, so it reduces overshoot and steadies the system ✓✓. (6)

3.4



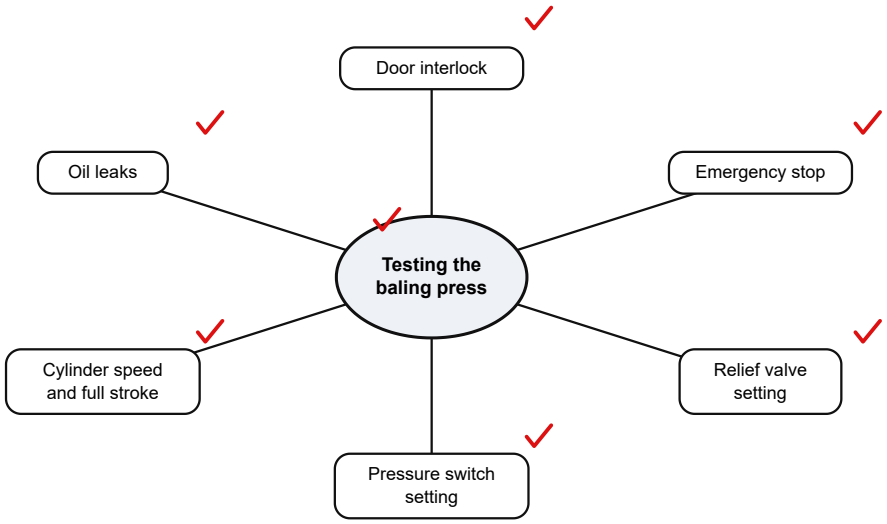
Inputs ✓ (START, STOP, E-stop, door switch, pressure switch, top limit switch); controller (PLC) ✓; outputs ✓ (pump motor, valve solenoids Y1 and Y2, lamps / alarm); power supply ✓; arrows showing the direction of signals ✓; labels ✓ (6)

[30]

QUESTION 4

4.1 It limits the highest pressure in the system ✓, so the cylinder, hoses and frame are protected from damage or bursting ✓. (2)

4.2



Centre topic "Testing the baling press" ✓ with any FIVE areas ✓✓✓✓✓: door interlock; E-stop; relief valve setting; pressure switch setting; cylinder speed and full stroke; oil leaks; pump motor direction and noise; bale quality. (6)

4.3 Start with a low setting ✓; compress a test bale while watching the gauge ✓; raise the setting in small steps until the bale is tight at the required pressure (below the relief setting), then lock the adjustment ✓. (3)

4.4 To check the movements, sequence and interlocks safely ✓ without risk to the material, the machine or people ✓. (2)

4.5 Any TWO ✓✓: safety glasses / face shield; gloves; safety boots; overalls; hearing protection. (2)

[15]

QUESTION 5

5.1

Task	How often
Check the oil level and look for leaks ✓	Daily
Check the door interlock and E-stop ✓	Daily
Clean or replace the return filter ✓	Every 3 months
Inspect hoses and fittings for wear ✓	Monthly
Change the hydraulic oil ✓	Yearly

Any FOUR sensible tasks ✓✓✓✓ with a sensible frequency for each ✓✓✓✓ (8)

5.2 Any THREE ✓✓✓: relief valve set too low so oil keeps dumping; oil level too low; blocked or dirty cooler / filter; wrong oil grade; worn pump; system working at high pressure for long periods. (3)

5.3 Lower the platen to rest / support it mechanically ✓; switch off and lock out the electrical supply ✓; release the stored pressure in the system (and accumulator) ✓; put up a warning tag and test that the press cannot move ✓. (4)

5.4 Oil leaking at high pressure can be injected through the skin ✓, causing serious injury or poisoning; use a piece of cardboard to find the leak ✓. (2)

5.5 Oil samples are taken at regular intervals and tested in a laboratory ✓ for wear metals, water and dirt ✓, so worn parts can be found and replaced before they fail ✓. (3)

[20]

TOTAL: 100

END OF MARKING GUIDELINE